

Working Concept Of This Model

PROJECT DOCUMENTATION

Title:

Smart Product Recommendation System using Collaborative Filtering

1. INTRODUCTION

A **Recommendation System** is a data-driven application that suggests items (products, movies, music) to users based on their preferences and past behavior.

It is widely used in platforms like:

- E-commerce (Amazon)
- Streaming (Netflix, Spotify)
- Social Media (YouTube)

2. OBJECTIVE

- To analyze user behavior (ratings)
- To predict user preferences
- To recommend relevant items

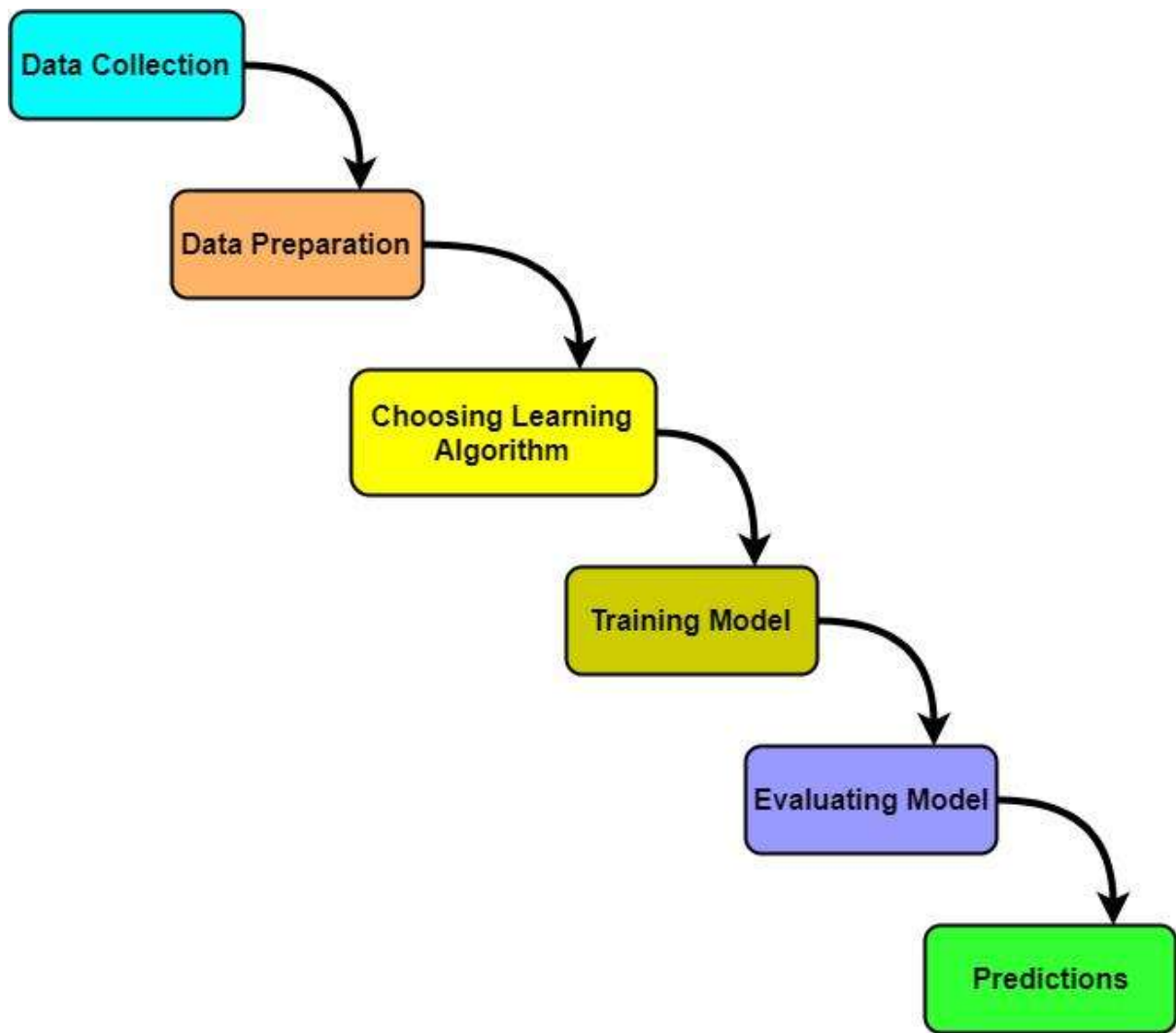
3. WORKING PRINCIPLE

This project uses **Collaborative Filtering**, which works on:

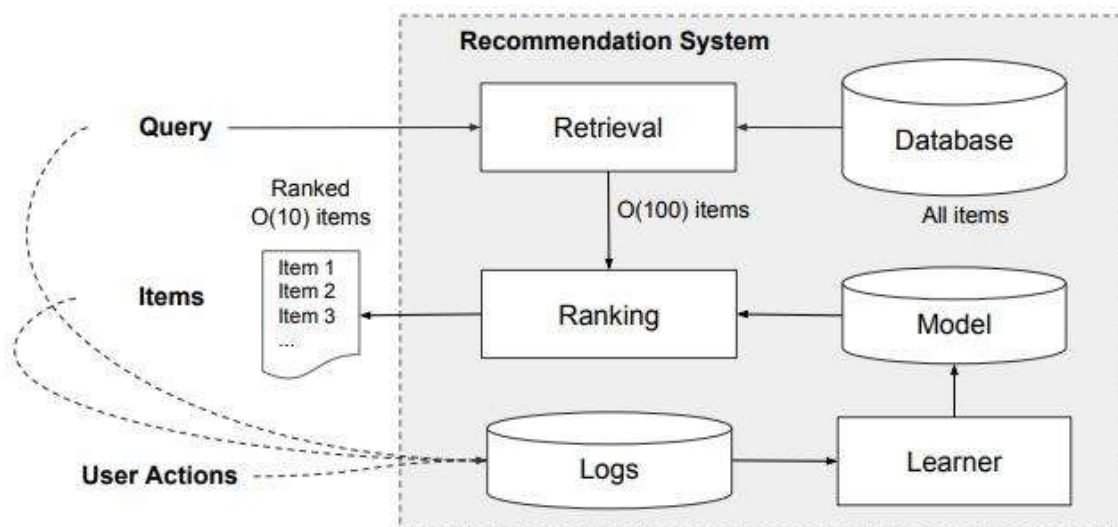
👉 “Users with similar behavior will like similar items”

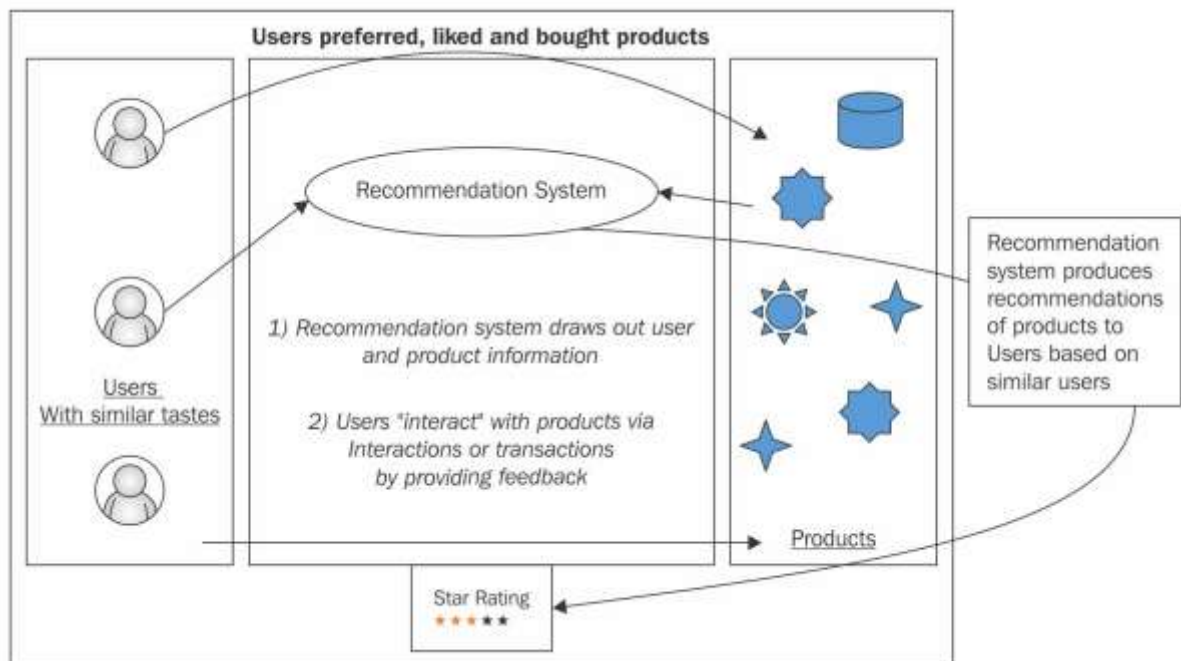
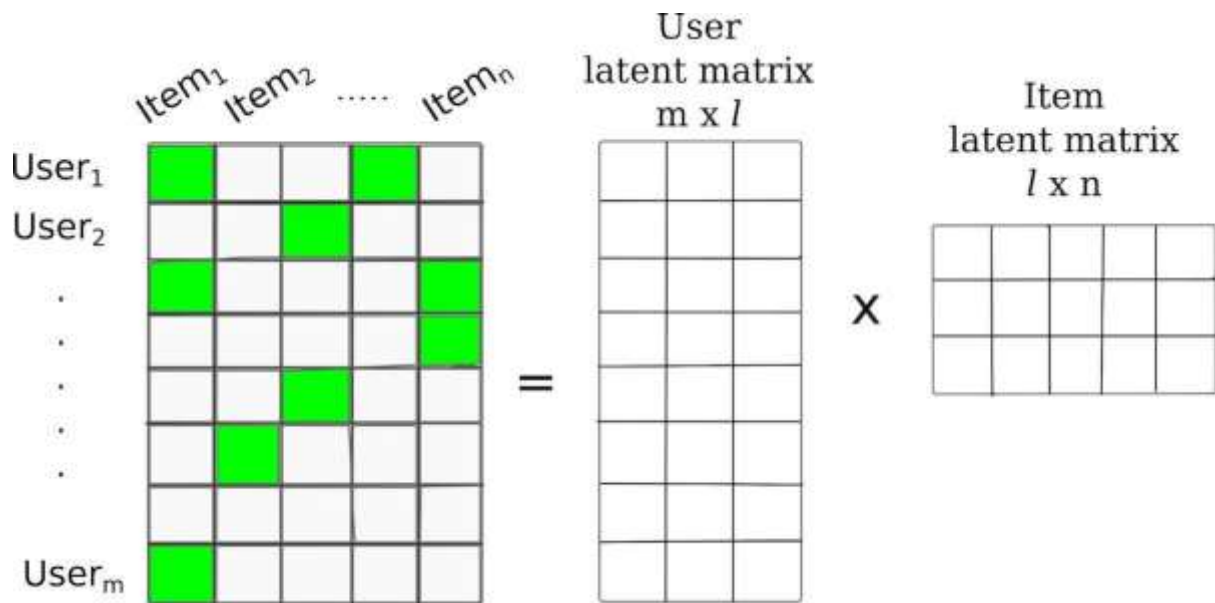
4. SYSTEM WORKFLOW

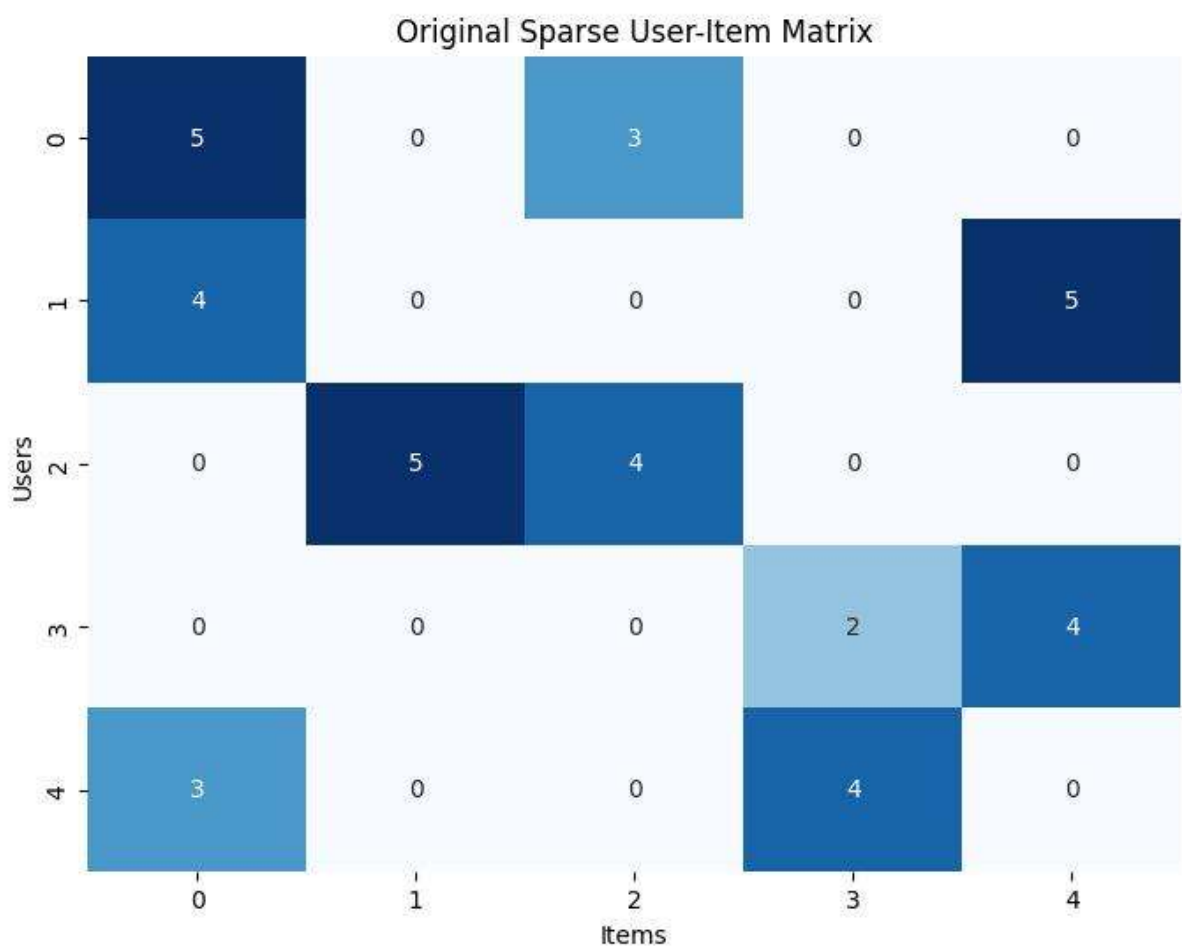
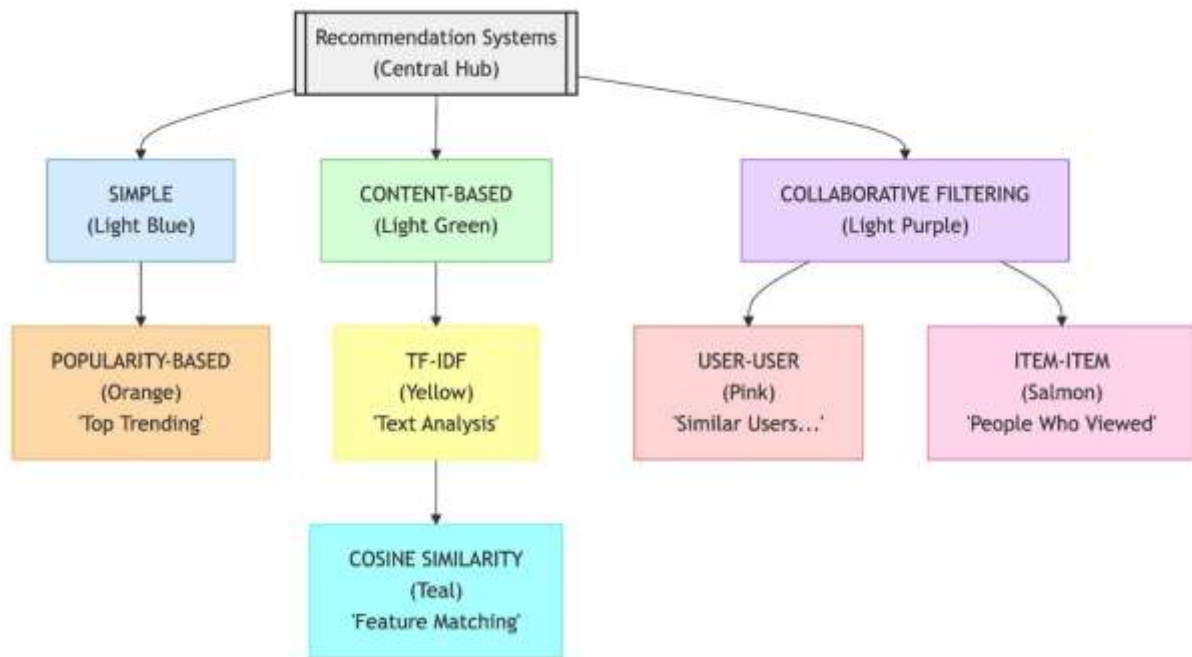
Step-by-Step Flow



Machine Learning Workflow







◆ Steps Explanation:

1. Data Collection

- UserId, ItemId, Rating collected

2. Data Preprocessing

- Clean missing values
- Format dataset

3. User-Item Matrix Creation

- Rows → Users
- Columns → Items
- Values → Ratings

4. Model Training (SVD)

- Learns hidden patterns (latent features)

5. Prediction

- Predict unknown ratings

6. Recommendation

- Suggest top-rated items

5. MATHEMATICAL CONCEPT (CORE IDEA)

The model predicts rating using:

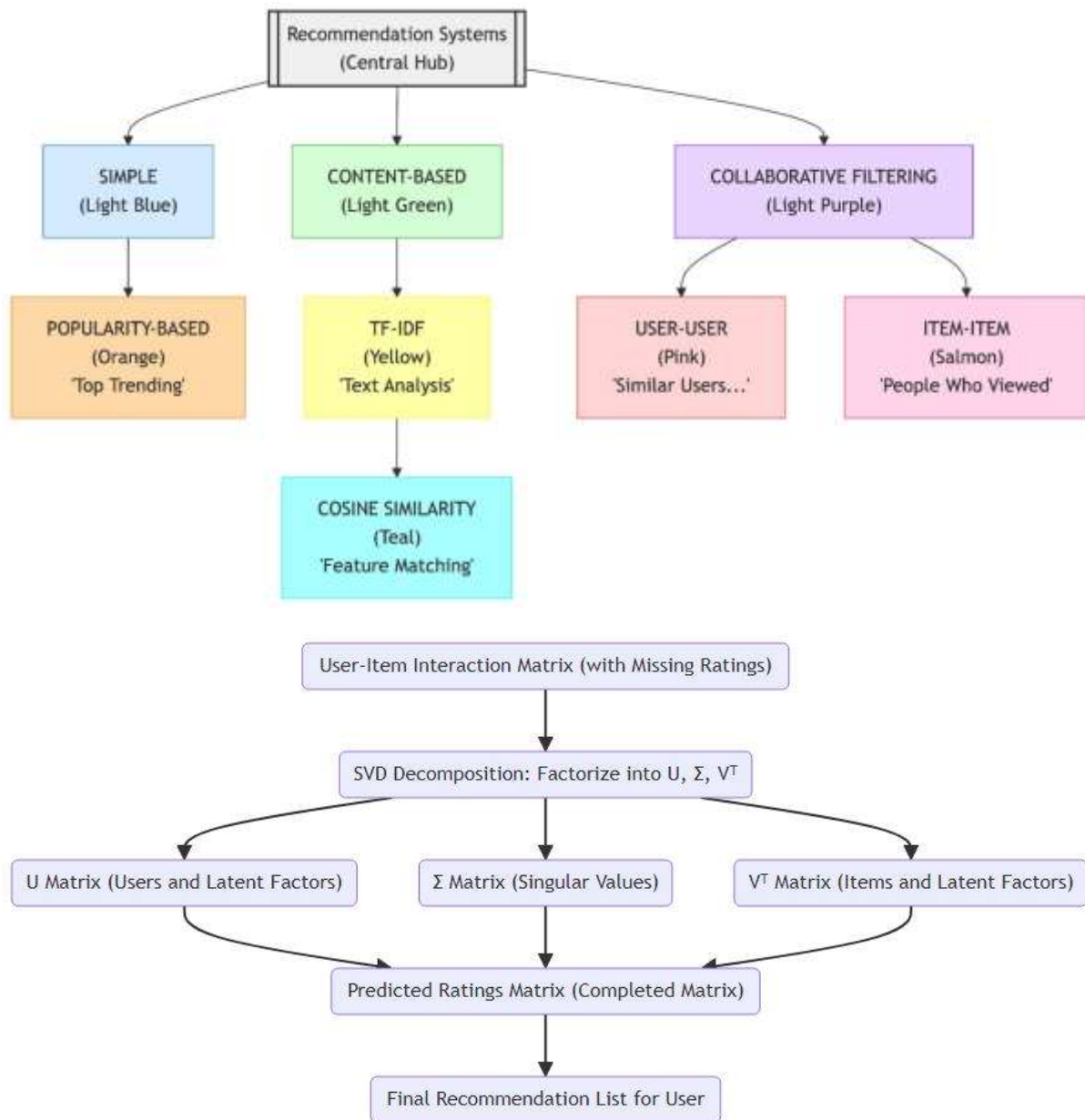
$$\hat{r}_{ui} = \mu + b_u + b_i + q_i^T p_u$$

Where:

- $(\hat{r}_{ui})$ = predicted rating
- (μ) = global average
- (b_u) = user bias
- (b_i) = item bias
- (p_u, q_i) = latent factors

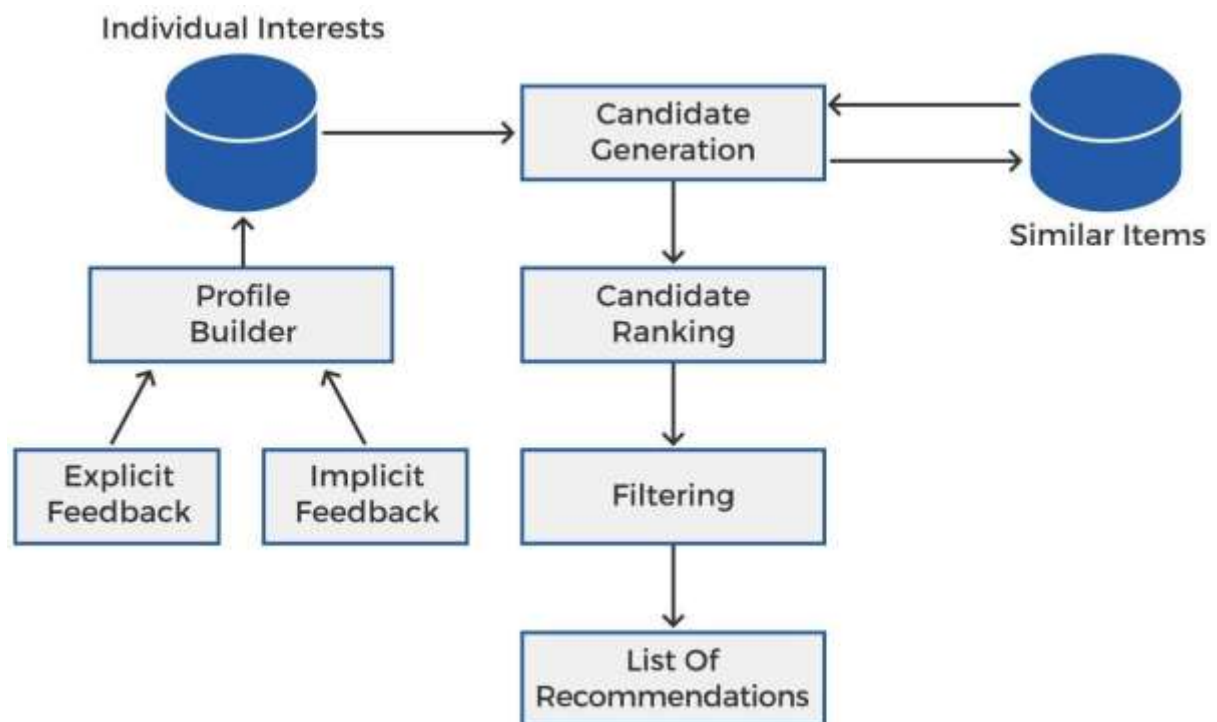
👉 This is used in **SVD (Matrix Factorization)**

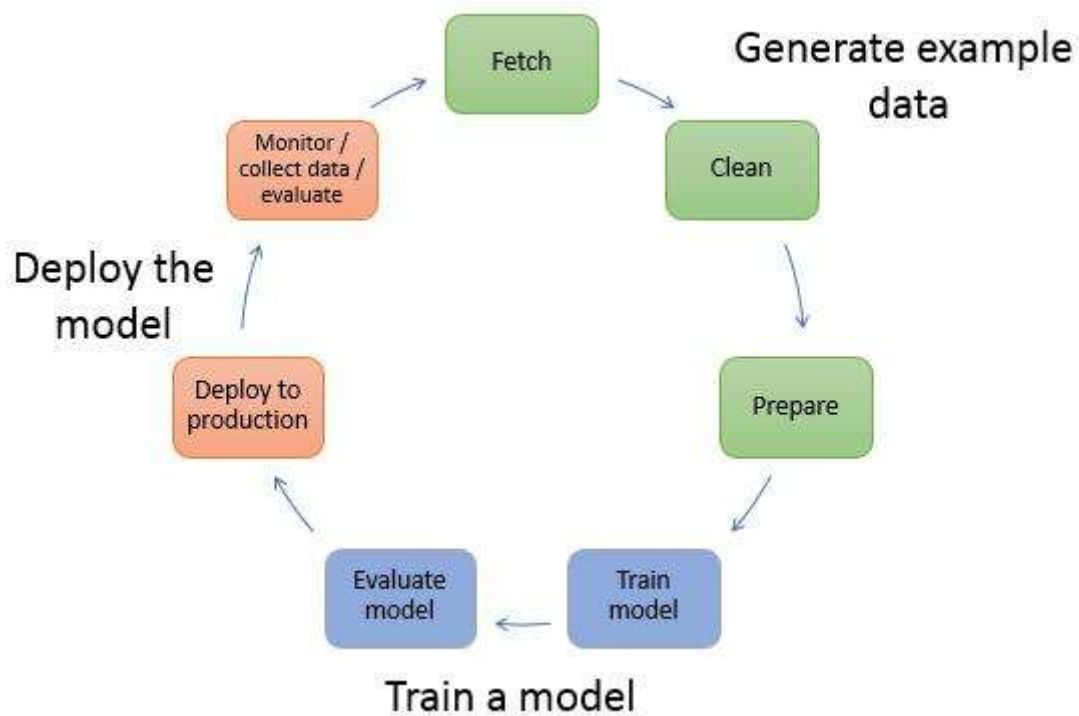
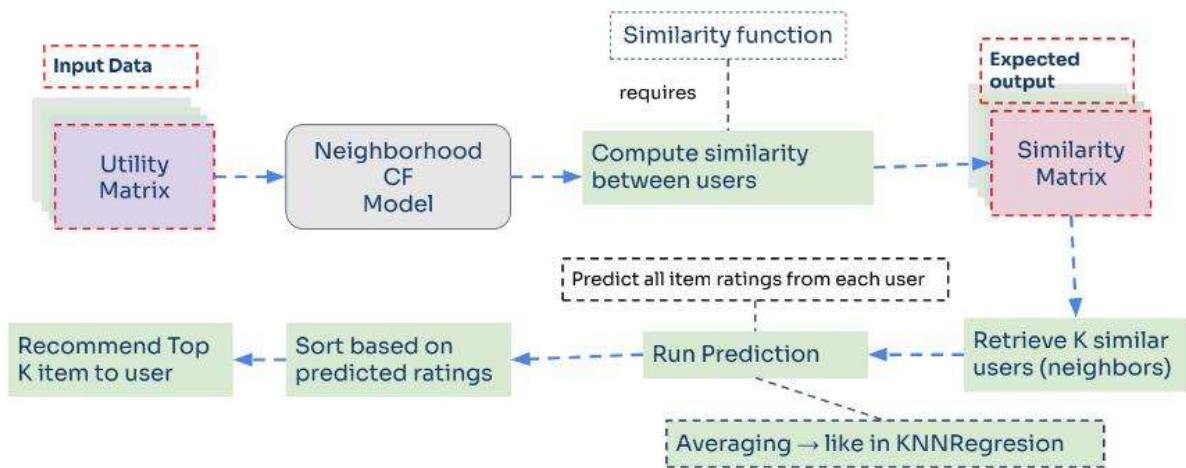
6. SYSTEM ARCHITECTURE



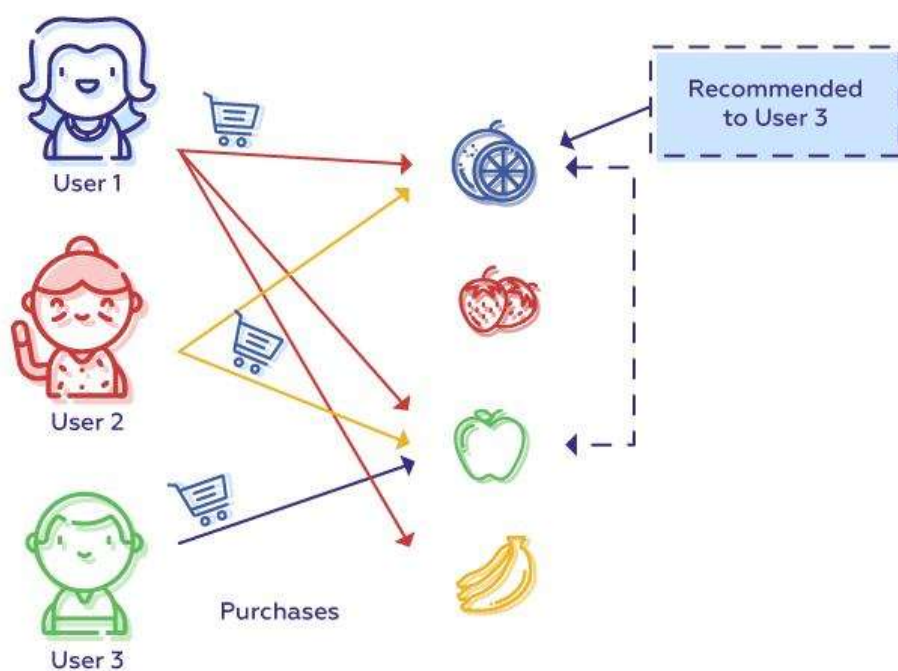
USER ITEM INTERACTION MATRIX

	ITEM 1	ITEM 2	ITEM 3	ITEM 4
USER 1	X		X	
USER 2		X		X
USER 3	X	X	X	X
USER 4	X	X		X
USER 5			X	
USER 6	X		X	X





Item-based filtering



sciforce

◆ Components:

- Input Layer → Dataset
- Processing Layer → Model Training
- Output Layer → Recommendations

🖥️ 7. IMPLEMENTATION (SCRIPTED EXPLANATION)

◆ Step 1: Import Libraries

```
import pandas as pd
```

```
from surprise import Dataset, Reader, SVD
```

◆ Step 2: Load Dataset

```
df = pd.read_csv("ratings.csv")
```

◆ Step 3: Convert to Surprise Format

```
reader = Reader(rating_scale=(1,5))
```

```
data = Dataset.load_from_df(df[['userId','itemId','rating']], reader)
```

◆ Step 4: Train Model

```
model = SVD()
```

```
model.fit(trainset)
```

◆ Step 5: Predict Rating

```
model.predict(user_id, item_id)
```

8. INPUT & OUTPUT

 **Input:**

userId	itemId	rating
1	101	4

 **Output:**

Predicted Rating: 4.6

Recommended Items: [102, 105, 110]

9. ADVANTAGES

- Personalized recommendations
- Improves user experience
- Increases sales/engagement

10. LIMITATIONS

- Cold start problem
- Sparse data issue

- Requires large dataset

11. REAL-WORLD APPLICATIONS

- Amazon → Product suggestions
- Netflix → Movie recommendations
- Spotify → Music playlists

12. FUTURE ENHANCEMENT

- Hybrid recommendation system
- Deep learning models
- Real-time recommendations

13. CONCLUSION

This project successfully demonstrates how machine learning can be used to build intelligent recommendation systems using collaborative filtering techniques like SVD.